

Automated Backup System using AWS Lambda and Amazon S3

Project Overview :

This project demonstrates how to build an automated backup system in AWS using AWS Lambda and Amazon S3. The system automatically creates backups of important files or data and stores them securely in an S3 bucket.

Automation reduces manual work, improves reliability, and ensures that backups happen on a scheduled basis.

Architecture :

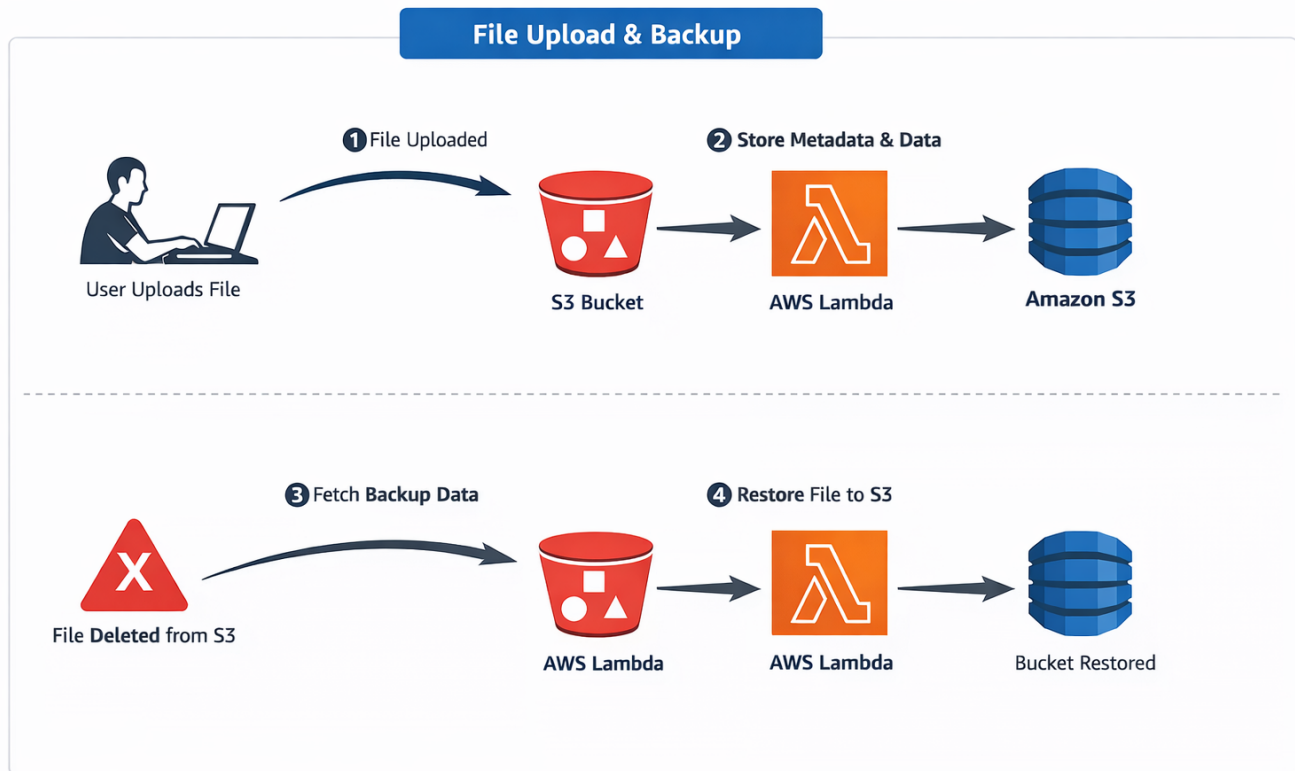
The backup workflow works as follows:

1. A scheduled trigger (Amazon EventBridge / CloudWatch Events) invokes the Lambda function.
2. The Lambda function processes the backup logic.
3. The data or files are copied to an Amazon S3 bucket.
4. The S3 bucket stores the backup securely and can be used for recovery when needed.

AWS Services Used:

- AWS Lambda – Executes the backup script automatically.
- Amazon S3 – Stores the backup files.
- Amazon EventBridge / CloudWatch Events – Triggers the Lambda function on a schedule.
- IAM – Provides permissions for Lambda to access S3.

Project Architecture Diagram :

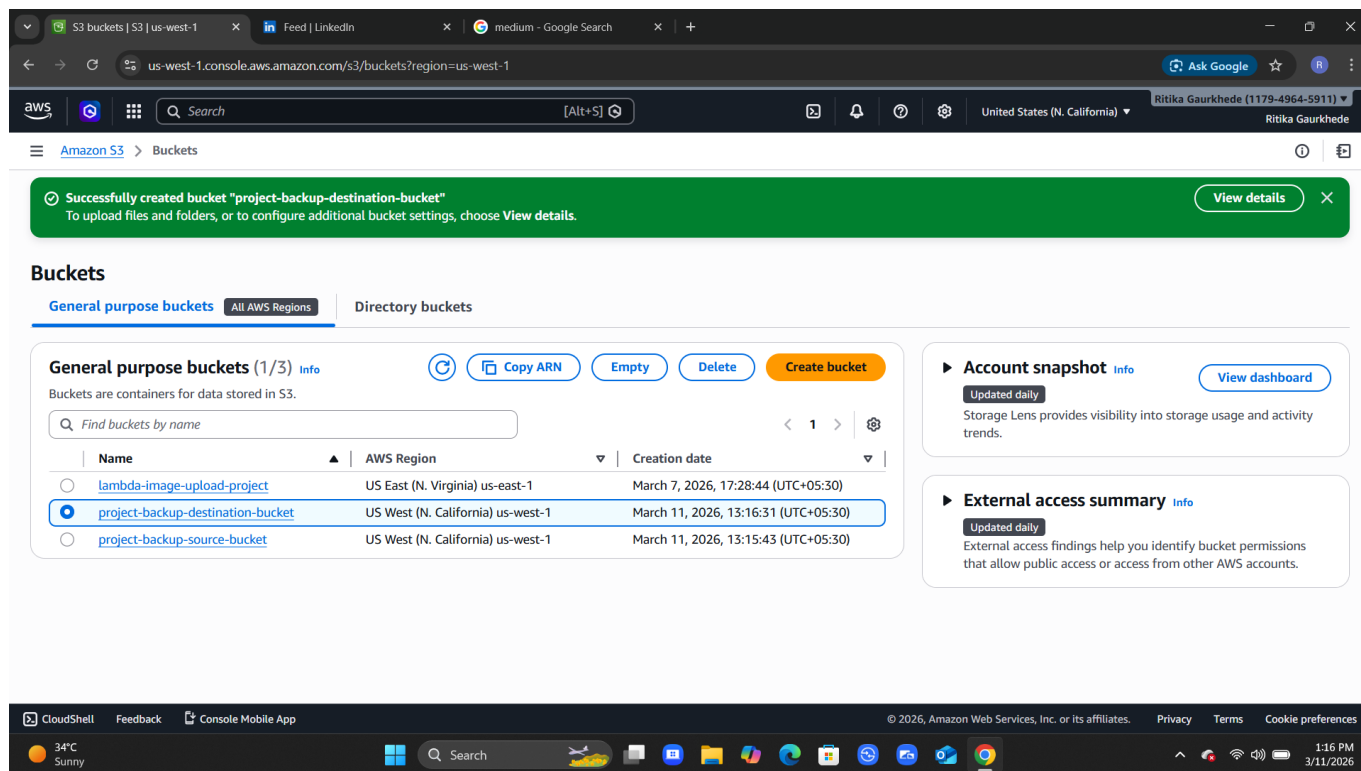


Automated S3 Backup & Restore System using AWS Lambda and Amazon S3

Project Setup Steps:

1. Create 2 S3 Bucket :

- Go to AWS Management Console.
- Open Amazon S3.
- Click Create bucket.
- Enter a unique bucket name.
- Select region.
- Create the bucket.



us-west-1.console.aws.amazon.com/s3/buckets?region=us-west-1

Successfully created bucket "project-backup-destination-bucket"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1/3) [Info](#) [Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Name	AWS Region	Creation date
lambda-image-upload-project	US East (N. Virginia) us-east-1	March 7, 2026, 17:28:44 (UTC+05:30)
project-backup-destination-bucket	US West (N. California) us-west-1	March 11, 2026, 13:16:31 (UTC+05:30)
project-backup-source-bucket	US West (N. California) us-west-1	March 11, 2026, 13:15:43 (UTC+05:30)

Account snapshot [Info](#) [View dashboard](#)
Updated daily
Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 34°C Sunny 1:16 PM 3/11/2026

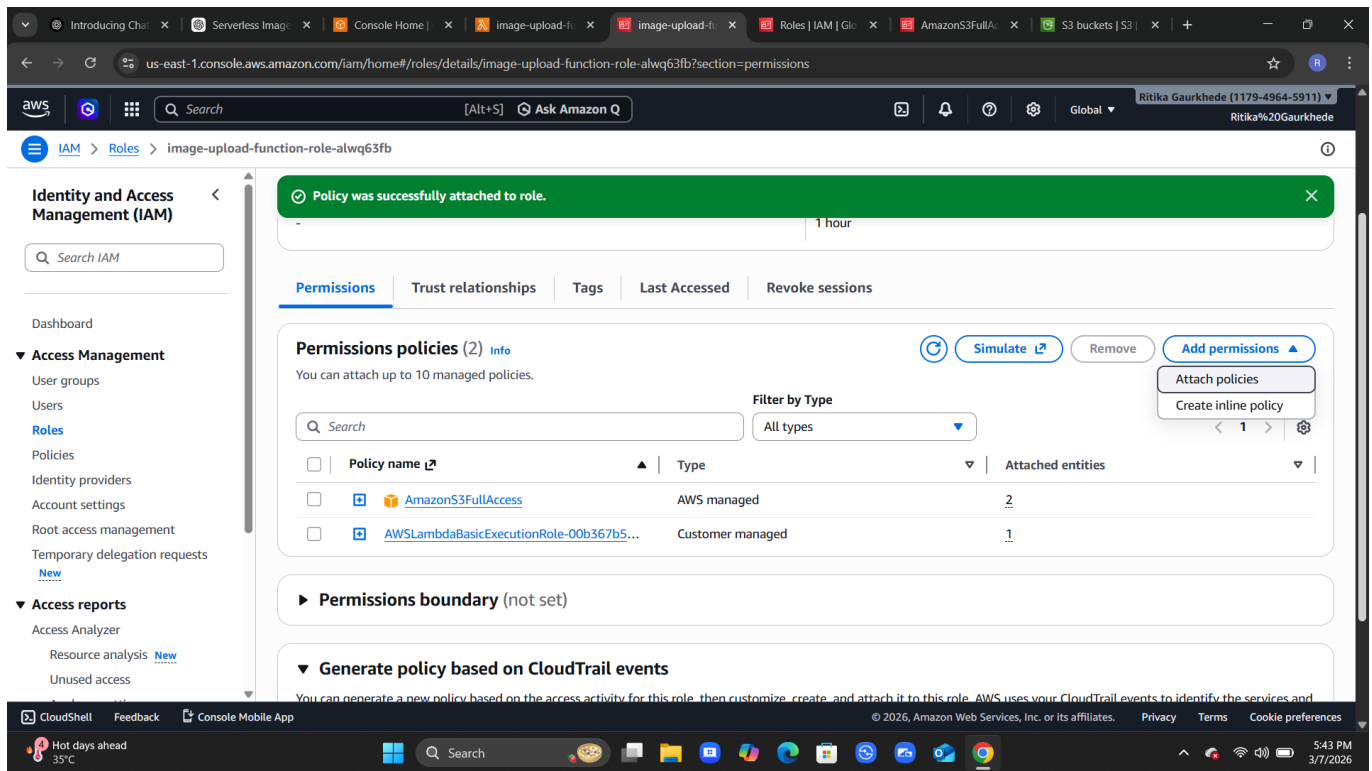
2. Create IAM Role for Lambda :

- Open IAM Console.
- Click Roles → Create Role.
- Select AWS Service → Lambda.

Attach policies:

AmazonS3FullAccess (or limited S3 access policy)

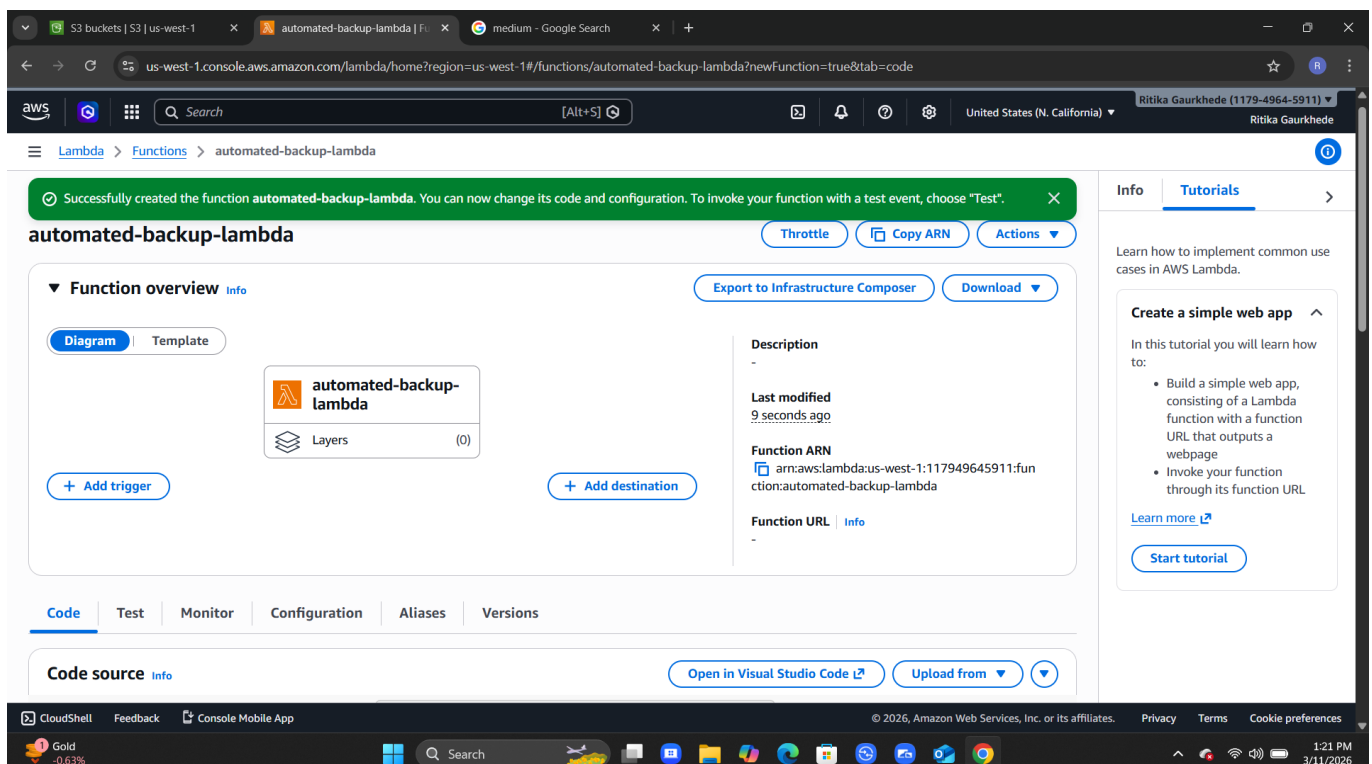
AWSLambdaBasicExecutionRole



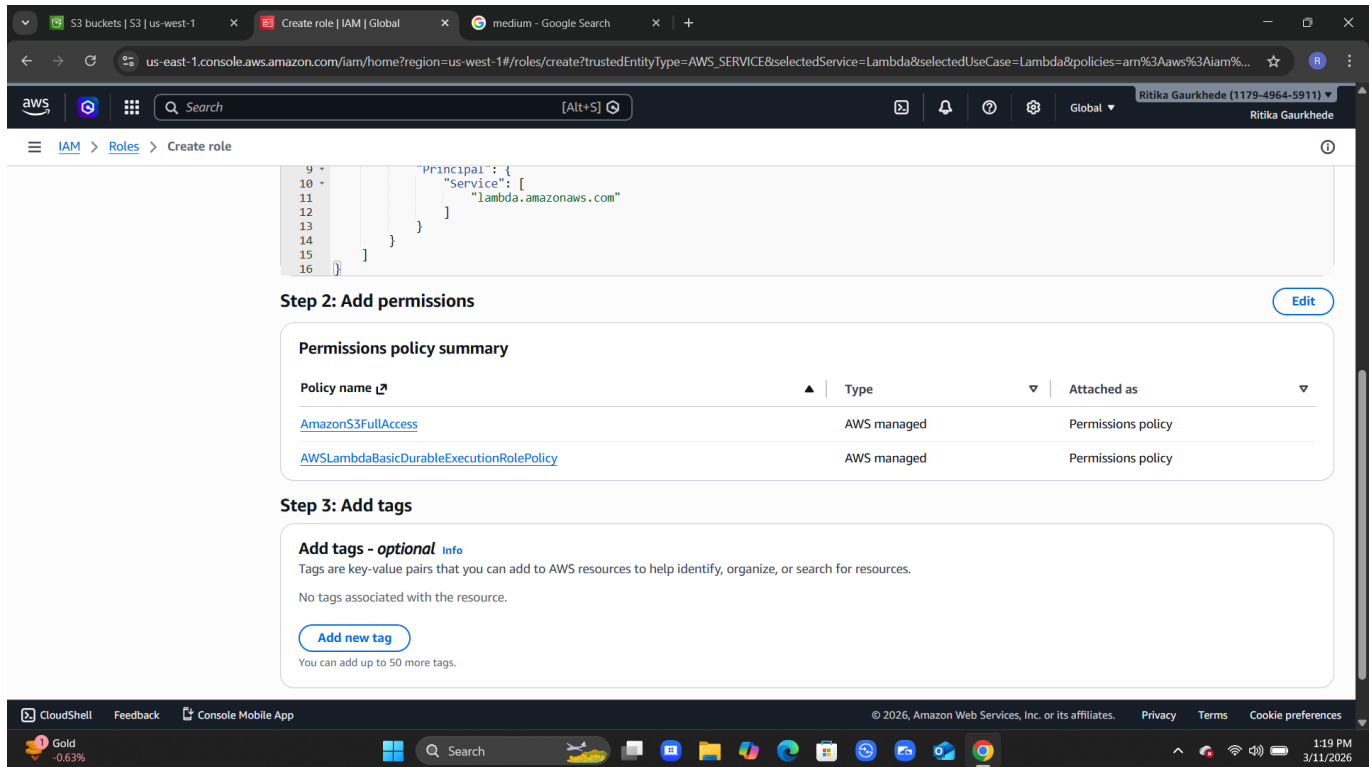
Create the role.

3. Create Lambda Function:

- Open AWS Lambda.
- Click Create Function.
- Select Author from scratch.
- Runtime: Python 3.12



Attach the IAM role created earlier.



Example Lambda Code:

```
import boto3
import datetime

s3 = boto3.client('s3')

source_bucket = 'project-backup-source-bucket'
destination_bucket = 'project-backup-destination-bucket'

def lambda_handler(event, context):

    response = s3.list_objects_v2(Bucket=source_bucket)

    if 'Contents' in response:
        for file in response['Contents']:
            file_key = file['Key']

            copy_source = {
                'Bucket': source_bucket,
                'Key': file_key
            }

            timestamp = datetime.datetime.now().strftime("%Y-%m-%d-%H-%M-%S")

            backup_key = f"backup-{timestamp}-{file_key}"

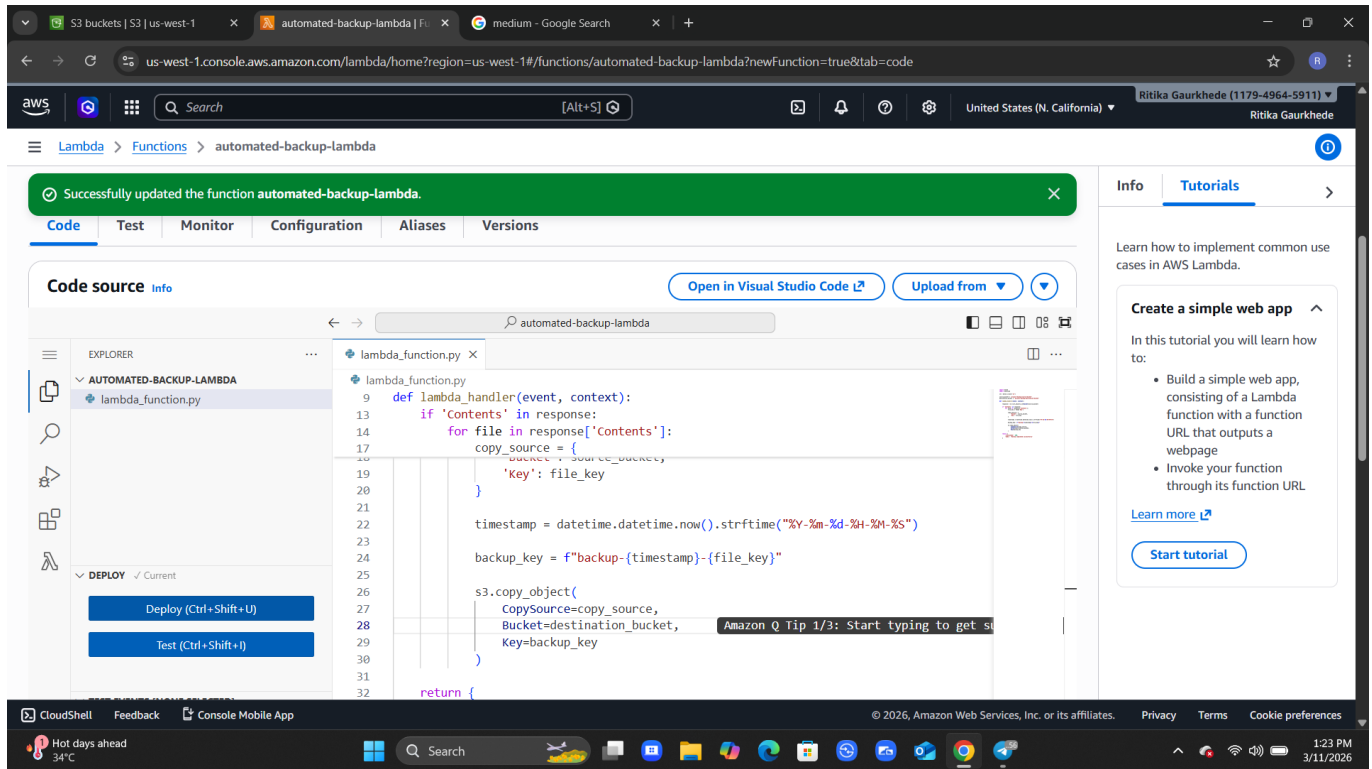
            s3.copy_object(
                CopySource=copy_source,
                Bucket=destination_bucket,
```

```

        Key=backup_key
    )

    return {
        'statusCode': 200,
        'body': 'Backup completed successfully'
    }

```

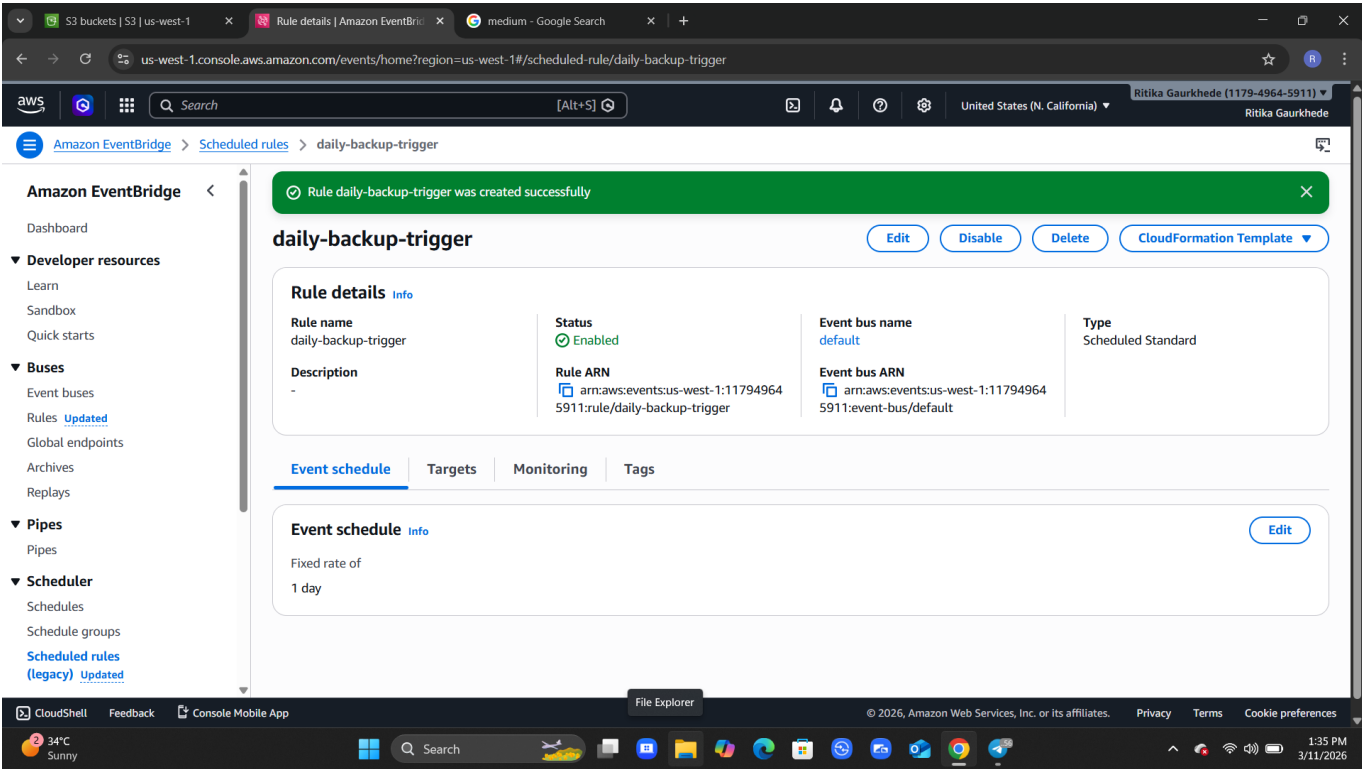


4. Create Scheduled Trigger:

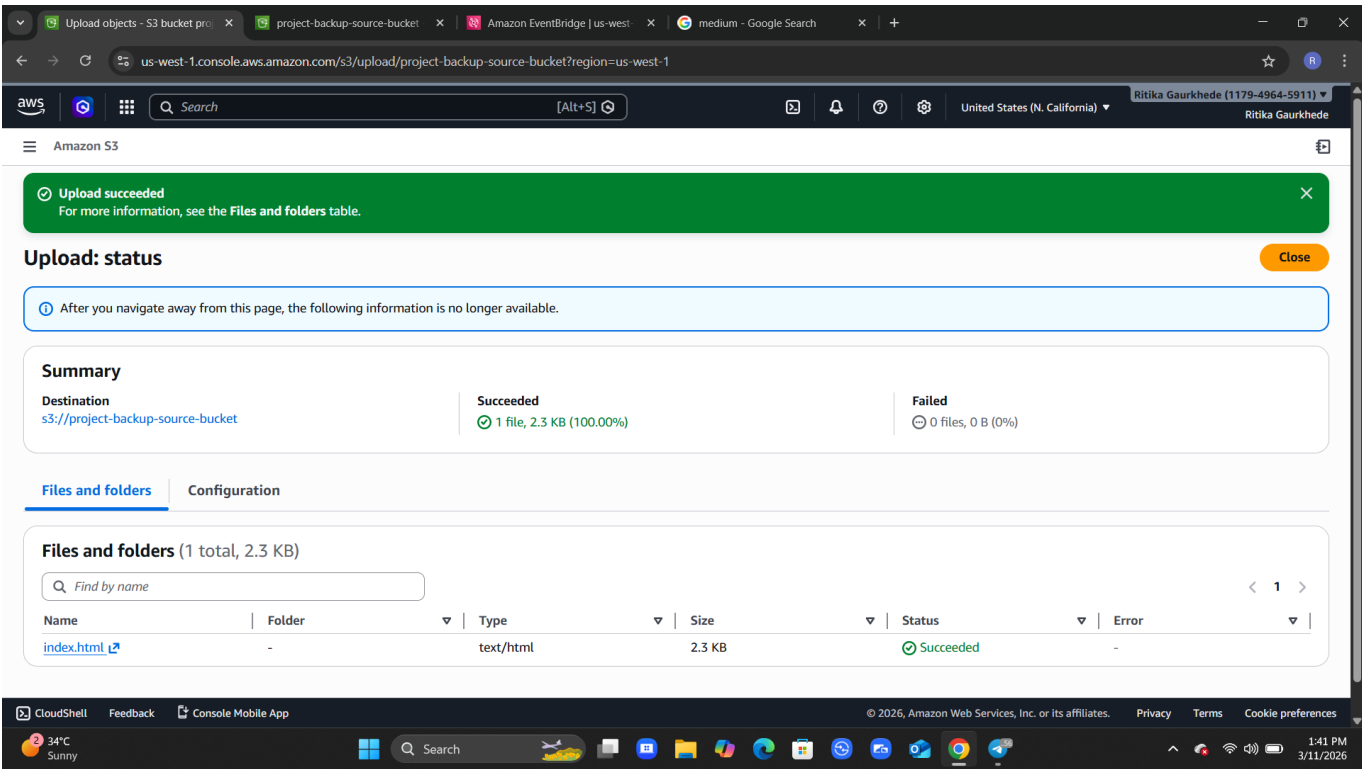
- Open Amazon EventBridge.
- Click Create Rule.
- Choose Schedule Pattern.

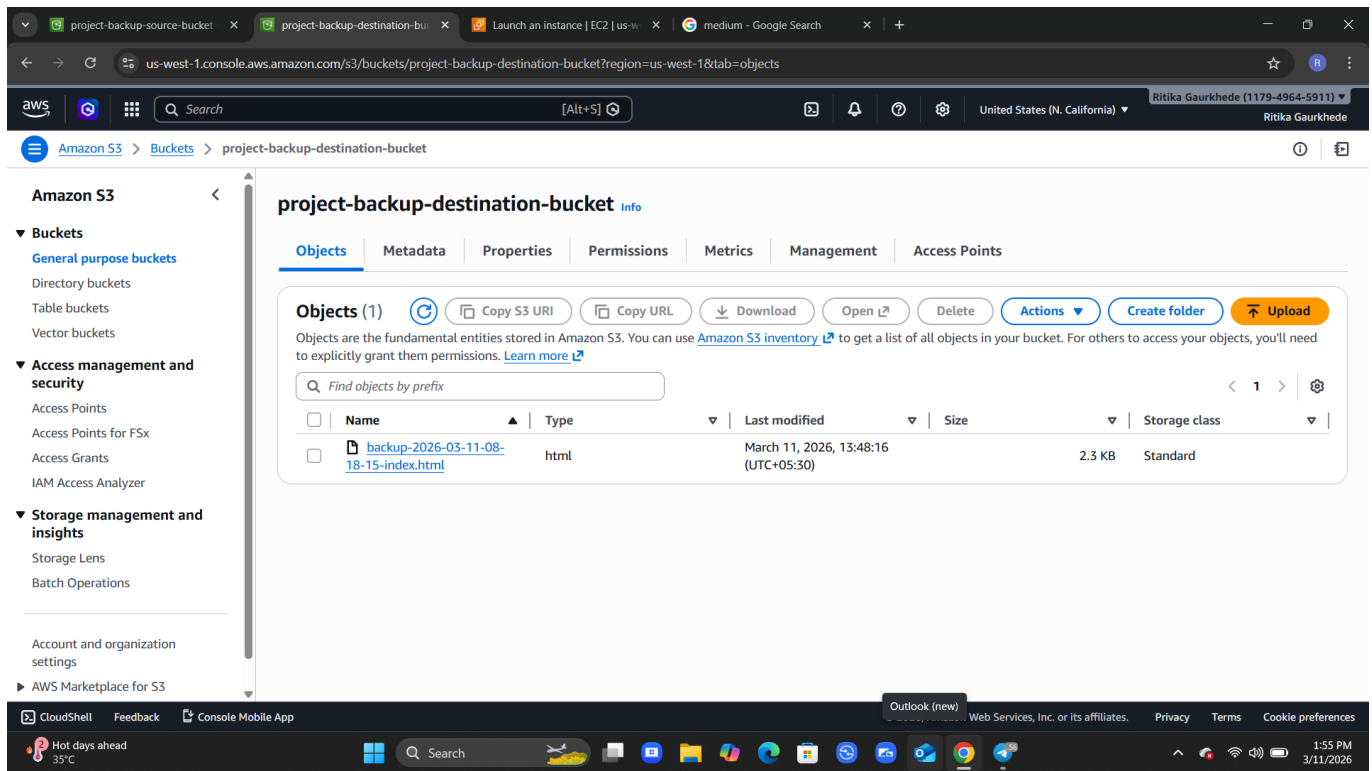
Example schedule: Run every day

- Select Lambda function as target.



This will automatically run the backup process.





Project Features:

1. Automated backups
2. Serverless architecture
3. Secure cloud storage
4. Scheduled backup execution
5. Scalable and cost-effective

Use Cases:

1. Website file backups
2. Application log backups
3. Database export backups
4. Disaster recovery preparation

Conclusion:

Automating backups is an essential part of modern cloud infrastructure. Using AWS services like Lambda and S3, we can build a reliable backup system without managing servers.

This project demonstrates how serverless automation can simplify infrastructure management while improving data safety.